Project: Worked examples
Title: Thermal expansion of a restrained bar
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the free movement, restraint force, axial stress and actual movement of a bar heated or cooled between two restraints. A gap and a rigid or elastic restraint are allowed, and the stress is compared with an allowable.

WHAT IT CALCULATES	RELATION
Free expansion of the bar	$\delta_{free} = \alpha \cdot L \cdot \Delta T$
Expansion actually restrained, once any gap g has closed	$\delta_e = \text{sign}(\delta_{free}) \cdot \max(\lvert\delta_{free}\rvert - g, 0)$
Axial stiffness of the bar	$k_{bar} = E \cdot A / L$
Restraint force, from rigid supports or an elastic restraint k_s	$F = \delta_e \cdot k_{bar}$ (rigid); $F = \delta_e / (1/k_{bar} + 1/k_s)$ (elastic)
Thermal stress in the bar, compressive on heating	$\sigma = -F/A$
Net movement of the bar end	$\delta = \delta_{free} - F/k_{bar}$

Basis: Thermal stresses[1](Ch. 15, 8th ed.); Thermal stresses and strains[2](§3-17, 10th ed.).

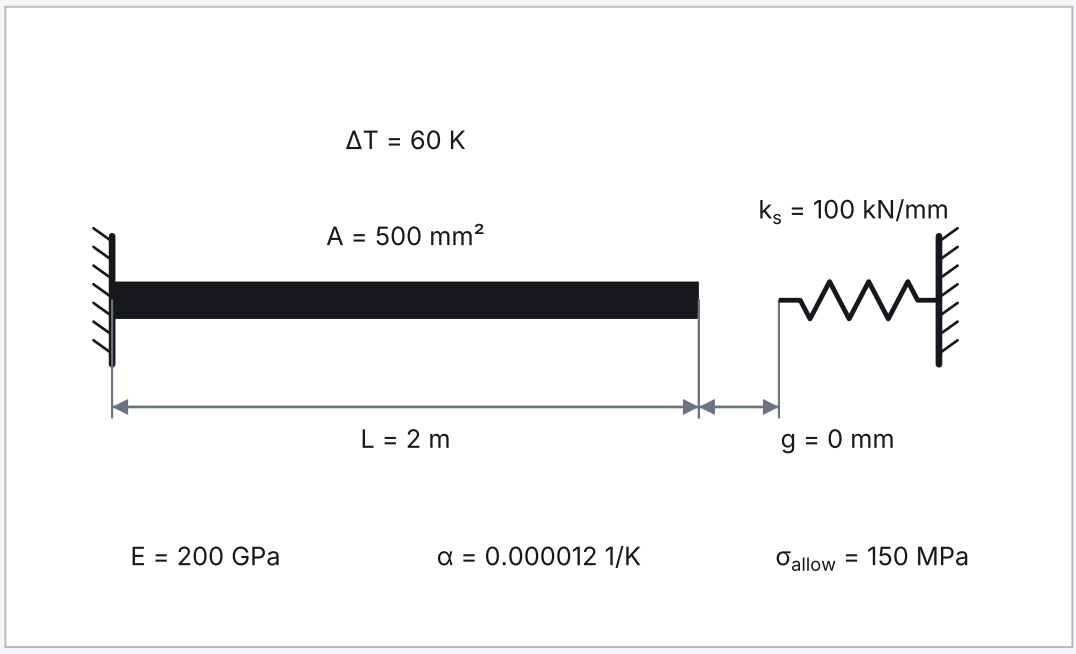


Figure 1 — A bar fixed at one end, a gap, then an elastic restraint of stiffness k_s. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
L	Bar length	2	m	drawing: DRG-TX-502 rev A	0.01 … 1000 m

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
A	Cross-section area	500	mm²	drawing: DRG-TX-502 rev A	1 … 1000000 mm²
g	Gap closed before the restraint engages	0	mm	drawing: DRG-TX-502 rev A	0 … 1000 mm
E	Young's modulus	200	GPa	drawing: DRG-TX-502 rev A	1 … 500 GPa
α	Coefficient of thermal expansion	0.000012	1/K	drawing: DRG-TX-502 rev A	1e-7 … 0.0001 1/K
dT	Temperature change (positive heating)	60	K	spec: LOAD-SCHEDULE-01	-300 … 1000 K
k_s	Restraint stiffness (optional; blank = a rigid restraint)	100	kN/mm	spec: SPEC-MECH-03 design criteria	0.001 … 10000000 kN/mm
σ_{allow}	Allowable stress	150	MPa	spec: SPEC-MECH-03 design criteria	1 … 2000 MPa

3. Assumptions

A1. The expansion coefficient is constant over the temperature change and the bar stays elastic. (Linear thermoelasticity.)
A2. The bar is uniformly heated and does not buckle under the compressive force. (Check buckling separately (struct.column-buckling).)

4. Working

[9] $\delta_{free} = \alpha * L * dT$ [2](§3-17)
= 0.000012 1/K × 2 m × 60 K
= 1.44 mm
Free thermal movement (positive on heating)

[10] $\delta_e = \delta_{free} - g$
= 1.44 mm − 0 mm
= 1.44 mm
Movement after the gap closes (engaged movement)

[11] $k_{bar} = E * A / L$
= 200 GPa × 500 mm² / 2 m
= 50000 kN/m
Axial stiffness of the bar

[12] $F = \delta_e / (1 / k_{bar} + 1 / k_s)$ [1](Ch. 15)
= 1.44 mm / (1 / 50000 kN/m + 1 / 100 kN/mm)
= 48 kN
Restraint force (elastic restraint in series with the bar)

[13] $\sigma = -F / A$
= −48 kN / 500 mm²
= -96 MPa
Axial stress (negative compressive)

[14] $\delta = \delta_{free} - F / k_{bar}$
= 1.44 mm − 48 kN / 50000 kN/m
= 0.48 mm
Actual movement

[15] σ_{abs} = abs(σ)
= abs(-96 MPa)
= 96 MPa
Stress magnitude
[16] u = σ_{abs} / σ_{allow}
= 96 MPa / 150 MPa
= 0.64
Stress utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
δ_{free}	1.440 mm	Free movement
F	48 kN	Restraint force
σ	-96 MPa	Axial stress
δ	0.4800 mm	Actual movement
δ_e	1.440 mm	Engaged movement
k_{bar}	50000 kN/m	Bar stiffness
σ_{abs}	96 MPa	Stress magnitude (governing)
u	0.64	Stress utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Thermal stress against allowable	96 MPa	150 MPa	0.64	

7. Applicability

None.

8. Verification

Template mech.thermal-expansion@1.1.1 Content hash 92ac49a0073a776b Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Young WC, Budynas RG, Sadegh AM. Roark's Formulas for Stress and Strain. 8th ed. McGraw-Hill; 2012.
2. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.